

9.1 Finding the Area of Two-Dimensional Figures

Lesson Introduction

 Benchmark	MA.912.GR.4.4 Solve mathematical and real-world problems involving the areas of two-dimensional figures.		 Learning Target	I can solve mathematical and real-world problems involving the area of two-dimensional figures.
 Benchmark Coherence	Building On		Working Toward	
	MA.6.GR.2.2 Solve mathematical and real-world problems involving the areas of quadrilaterals and composite figures by decomposing them into triangles or rectangles. MA.7.GR.1.1 Apply formulas to find the areas of trapezoids, parallelograms, and rhombi. MA.7.GR.1.4 Explore and apply a formula to find the area of a circle to solve mathematical and real-world problems.		MA.912.GR.3.4 Use coordinate geometry to solve mathematical and real-world problems on the coordinate plane involving perimeter or areas of polygons. MA.912.GR.4.3 Extend previous understanding of scale drawings and scale factors to determine how dilations affect the areas of two-dimensional figures and the surface areas or volumes of three-dimensional figures. MA.912.GR.4.6 Solve mathematical and real-world problems involving the surface areas of three-dimensional figures limited to cylinders, pyramids, prisms, cones, and spheres.	
 Essential Questions	- What is area? - How can certain quadrilaterals be divided into triangles and rectangles? - What are the formulas for area of a rectangle and area of a triangle?			
 Lesson Narrative	This lesson gives students an opportunity to review areas of triangles and rectangles. Students are given time to explore how to find the areas of various quadrilaterals by dividing the polygons into areas of triangles and rectangles. Students will then use an example of using the area of a circle to find the area of a semicircle. Students can derive formulas for a parallelogram, trapezoid, and rhombus by dividing the figures into rectangles and triangles.			
 Component Summary	9.1.1 Warm-Up (~5 min.)	Examining a quote from Michael Jordan for growth mindset (<i>SE p. 51</i>)	9.1.3 Guided Practice (5-10 min.)	Finding the areas of figures (<i>SE p. 56</i>)
	9.1.2 Collaboration (20-25 min.)	Exploring the areas of various polygons by decomposing (<i>SE p. 52</i>)	9.1.4 Your Turn! (5 min.)	Finding the area of a composite figure (<i>SE p. 58</i>)
	9.1.5 Wrap-Up (~5 min)	Reflecting on the level of understanding of finding the areas of various figures (<i>SE p. 58</i>)		
 Resources	Glossary		Materials	
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> N/A		- Calculator (Optional) Colored Pencils/Pens	
			Additional Preparation	
			N/A	

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may struggle differentiating between the radius and diameter of a circle. - Students may misread the graphs in question 6 of 9.1.2 Collaboration and question 1 of 9.1.4 Your Turn! as the scale is in units of 0.5 instead of 1. - Students may not notice that in question 1 of 9.1.4 Your Turn! each 1 unit = 5 feet. - Students may forget to divide by 2 when finding the area of a triangle.	Consider allowing students to use tools, such as colored pencils and/or pens to divide the given quadrilaterals into rectangles and triangles so that the visuals might be more accessible for some students.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Discussion Supports 9.1.2 Collaboration lends itself to discussion about the formulas for areas of various figures. Allow students the opportunity to discuss the relationships between the formulas and the figures of the two-dimensional figures and give opportunities for students to construct their own knowledge using discussion supports.	MA.K12.MTR.5.1: Patterns and Structure 9.1.2 Collaboration gives students the opportunity to explore and discover how using the areas of rectangles and triangles can be applied to find the areas of other two-dimensional figures. Use MA.K12.MTR.5.1 as an opportunity for students to look at patterns and structures to be able to find the area of multiple two-dimensional figures.
 Differentiate Instruction	In 9.1.2 Collaboration, there are opportunities to differentiate with Discussion Supports. The following examples of sentence stems and frames could be used:	
	Sentence Stems: - One difference between the formula for the area of a rectangle and the area of a triangle is... - I can divide the parallelogram into two triangles by... - I can find the area of a rhombus by... Sentence Frames: Sentence frames allow all students an opportunity to approach the discussions. The underlined words are the words that would be provided for students to fill in the blanks. See the examples below: - The <u>radius</u> is <u>half</u> of the <u>diameter</u>. - The formula for area of a triangle is <u>one half</u> of the <u>base</u> multiplied by the <u>height</u>. - The formula for area of a rectangle is the <u>base</u> multiplied by the <u>height</u>.	
Lesson Notes:		

9.1.1 Warm-Up: Michael Jordan

Component Overview: Students begin the lesson by analyzing a quote from Michael Jordan about how failure leads to success. This gives students an opportunity to think about instances where failures can be turned into successes and how students can grow from failures.



9.1.1 Warm-Up: Michael Jordan

Basketball legend Michael Jordan is often quoted as saying, "I've failed over and over and over again in my life and that is why I succeed."

1. What is a skill, activity, or subject that you have failed at?

Answers will vary.

2. Explain Jordan's quote in your own words.

Answers will vary.

3. How can you turn a failure into a success story?

Answers will vary.



If time permits, prompt one to two students to share their responses on how failures lead to success. Also, give students an opportunity to describe their interpretations of the quote or offer extra credit for an extension project in which students write a response on how failure leads to success in Math class.

9.1.2 Collaboration: Finding Area

Component Overview: Students will begin working with the formulas for the area of a triangle and the area of a rectangle, which most students should be able to recall. Students will work with the formulas for the areas of a triangle and rectangle to derive the formulas for the areas a parallelogram, square, trapezoid, and rhombus. 9.1.2 Collaboration also gives students a quick opportunity to review and work with the formula for the area of a circle. For question 4, students may need some additional support and guidance on how to use the triangle to find the area of the parallelogram.



9.1.2 Collaboration: Finding Area

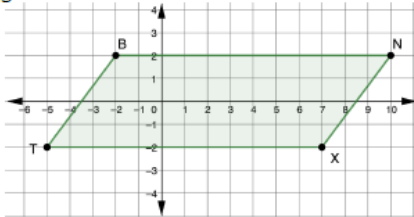
- Complete the table.

Figure	Area Formula
Rectangle	$A = \text{base} \times \text{height}$
Triangle	$A = \frac{1}{2} \text{base} \times \text{height}$

- How is the area of a triangle related to the area of a rectangle?

Both formulas use the measurements for base and height. The area of a triangle is $\frac{1}{2}$ the area of a rectangle with the same base and height measurements.

- Parallelogram $BNXT$ is shown on the coordinate grid.



Tyrese could not remember how to find the area of a parallelogram so he divided the parallelogram into 2 triangles and a rectangle. Work with your partner to use Tyrese's method to find the area of $BNXT$.

Shape	Area
Triangle 1	$A = \frac{4 \times 3}{2} = 6 \text{ units}^2$
Triangle 2	$A = \frac{4 \times 3}{2} = 6 \text{ units}^2$
Rectangle	$A = 4 \times 9 = 36 \text{ units}^2$
Total Area of Parallelogram	The total area of the shape would be the sum of the areas of the individual components. This would be $36 + 6 + 6 = 48 \text{ units}^2$.



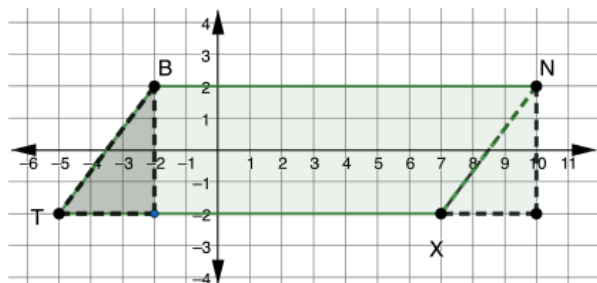
Why is the area of a triangle half the area of the rectangle? (Dividing a rectangle diagonally creates two right triangles.)



How did Tyrese know that dividing the parallelogram into two triangles and a rectangle would work? (The formula for the area of two triangles is equal to the formula for the area of a rectangle.)

9.1.2 Collaboration: Finding Area (continued)

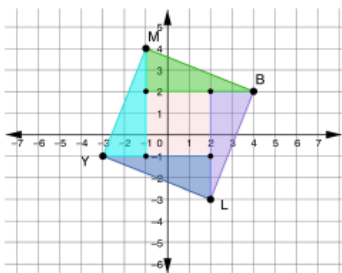
4. Sarianna rearranged the parallelogram to create a rectangle.



Use Sarianna's rectangle to find the area of $BNXT$.

The side lengths of the rectangle are 4 and 12. Thus the area is $4 \times 12 = 48 \text{ units}^2$.

5. Quadrilateral $MBLY$ is decomposed as shown on the coordinate grid. Find the area of $MBLY$ using the decomposition shown.



Area of Triangles = $4\left(\frac{2 \times 5}{2}\right) = 20 \text{ units}^2$

Square = $3 \times 3 = 9 \text{ units}^2$

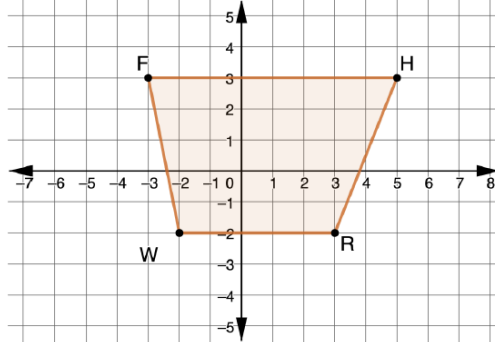
Total = $9 + 20 = 29 \text{ units}^2$



Consider the use of digital applets, such as GeoGebra to visually show the translation of the parallelogram.

9.1.2 Collaboration: Finding Area (continued)

6. $FHRW$ is shown on the coordinate grid.



- a. Divide the quadrilateral into two triangles and a rectangle. Then, find the area of each.

Shape	Area
Triangle 1	$A = \frac{5 \times 1}{2} = 2.5 \text{ units}^2$
Triangle 2	$A = \frac{5 \times 2}{2} = 5 \text{ units}^2$
Rectangle	$5 \times 5 = 25 \text{ units}^2$

- b. What is the area of $FHRW$?

The area of $FHRW$ is $2.5 + 5 + 25 = 32.5 \text{ units}^2$.

- c. What type of quadrilateral is $FHRW$?

$FHRW$ is a trapezoid.

7. Ava found a formula for the area of a trapezoid. The formula is $A = \frac{1}{2}h(b_1 + b_2)$, where h is the height and b_1 and b_2 are the bases. Use the formula Ava found to find the area of $FHRW$.

$$A = \frac{1}{2}(5)(5 + 8) = 32.5 \text{ units}^2.$$



For a visual approach, have students outline each triangle and the square to be able to better identify each individual figure, especially for students who may have trouble identifying various colors due to vision issues.



To increase the rigor, have students speculate which specific type of quadrilateral is graphed. Have students justify their answers mathematically. The specifics on identifying quadrilaterals on the coordinate plane will be taught in Unit 13.



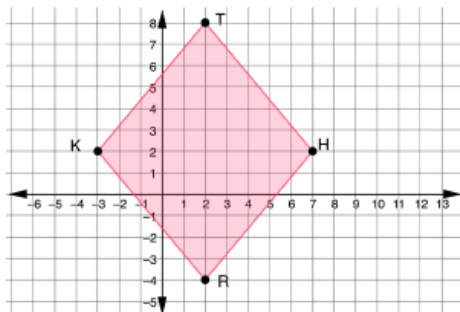
For a visual approach, have students use two separate colored pencils or pens to draw the two triangles on the coordinate plane.



How do you know that this quadrilateral is a trapezoid? (The quadrilateral has one set of parallel sides and two slanted sides.)

9.1.2 Collaboration: Finding Area (continued)

8. $THRK$ is shown on the coordinate grid.



a. What type of quadrilateral is $THRK$?

$THRK$ is a rhombus.

b. Determine the area of $THRK$. Show your work or explain your thinking.

Answers will vary. One way to determine the area of a rhombus is to draw diagonals and create 4 congruent right triangles, because diagonals bisect each other and are perpendicular in a rhombus. Each right triangle has an area of 15 units^2 , so the area of $THRK$ is 60 units^2 .

c. Complete the table.

Line Segment	Length
$\overline{TR}$	12 units
$\overline{KH}$	10 units

The area of a rhombus can be found using the formula

$A = \frac{1}{2}pq$, where p and q are the lengths of the diagonals.

d. Use the formula to find the area of $THRK$.

$$A = \frac{1}{2} \times 12 \times 10 = 60 \text{ units}^2$$



Prompt students for verbal explanations and utilize inquiry to extend student thinking for increased rigor.

- How do you know that this quadrilateral is a rhombus? (The quadrilateral has four congruent sides, perpendicular diagonals, diagonals that bisect each other.)



Students may categorize quadrilateral $THRK$ as a rhombus. Both answers are correct since, by definition, a rhombus is also a kite.

9.1.2 Collaboration: Finding Area (continued)

Formulas for area can easily be forgotten, so it is always helpful to be able to break a figure down into triangles and rectangles in order to find the area. The exception is finding the area of a circle or a part of a circle.

9. What is the area of a circle?

$$A = \pi r^2$$

10. The drama program at school is building a new semicircular stage. The longest distance across the stage is 30 feet. What is the area of the new stage? Round to the nearest thousandth.



$$A = \frac{\pi(15^2)}{2} = 353.429ft^2$$



Students may need to be reminded of the differences between the radius and diameter. Students may also need to be reminded that if the diameter is given, divide by two to find the radius.

9.1.3 Guided Practice: Using the Formulas

Component Overview: Students will begin working to find the areas of various two-dimensional figures. Students will find the area of a circle in question 1. Students may divide the figures into triangles and rectangles on questions 2-3, or the formulas for the parallelogram and rhombus that were given in the lesson may be used. Students will be required to divide the figure in question 4 into multiple figures and will be required to find the area of the composite figure.



9.1.3 Guided Practice: Using the Formulas

1. The radius of a circle is 7 feet (ft). Find the area, in square feet, of the circle. Leave the answer in π form.

$$A = \pi \cdot 7^2 = 49\pi \text{ ft}^2$$

2. The base of the parallelogram is 22 inches (in.). The area of the parallelogram is 132 square inches. Find the height, in inches, of the parallelogram.

$$A = bh; 132 = 22h; h = 6 \text{ inches}$$

3. The diagonals of a rhombus are 40 centimeters (cm) and 18 centimeters. Find the area, in square centimeters, of the rhombus.

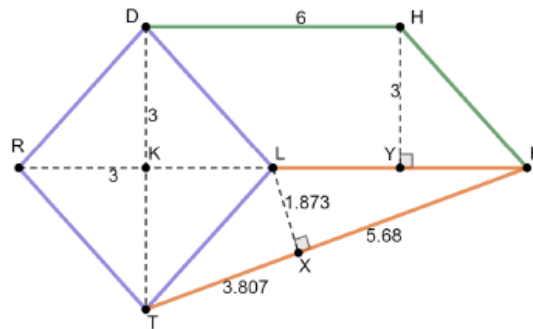
$$A = \frac{1}{2}(40 \times 18) = 360 \text{ cm}^2$$



Encourage students to sketch each figure, as students may be hiding underlying misconceptions if they just multiply the numbers in the problem. Sketching each figure provides insight into the students' grasps on concepts like radius vs. diameter (question 1), base vs. height (question 2), and diagonal vs. base (question 3).

9.1.3 Guided Practice: Using the Formulas (continued)

4. A composite shape of a rhombus, a parallelogram, and a triangle is shown.



The use of colored pencils or markers to clearly annotate the figures that construct the composite figure can be a helpful tool or strategy for students. Encourage students to decompose or mentally deconstruct each figure using the strategy that works best for them.



Students may see the side TL as the base of the triangle, with a height of 6 units, rather than what is depicted in the answer key.



Monitor for students who use 3 units instead of 6 units for the diagonal of the rhombus.

Complete the table. Round to the nearest thousandth.

Shape	Area
Rhombus	$A = \frac{1}{2}pq = \frac{6 \times 6}{2} = 18 \text{ units}^2$
Parallelogram	$A = bh = 3 \times 6 = 18 \text{ units}^2$
Triangle	$A = \frac{1}{2}bh = \frac{17.77}{2} = 8.885 \text{ units}^2$
Composite Area	$18 + 18 + 8.885 = 44.885 \text{ units}^2$

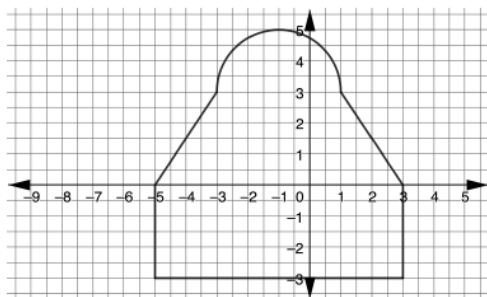
9.1.4 Your Turn!

Component Overview: Students will complete one question where dividing the composite figure into a semicircle, trapezoid, and rectangle will be required. Students must first complete this step before finding the area of the composite figure. Once the areas of the semicircle, trapezoid, and rectangle have been calculated, students must add to find the area of the composite figure.



9.1.4 Your Turn!

1. The design of a room is shown, where 1 unit = 5 feet.



Determine the area, in square feet, of the room. Round to the nearest hundredth.

Semi-circle ($r = 2 \text{ units} = 10 \text{ ft}$)

$$= \frac{\pi r^2}{2} = \frac{\pi(10^2)}{2} = \frac{100\pi}{2} = 50\pi = 157.080 \text{ ft}^2$$

Trapezoid ($h = 3 \text{ units} = 15 \text{ ft}$, $b_1 = 4 \text{ units} = 20 \text{ ft}$, $b_2 = 8 \text{ units} = 40 \text{ ft}$)

$$= \frac{15(20+40)}{2} = \frac{900}{2} = 450 \text{ ft}^2$$

Rectangle ($b = 8 \text{ units} = 40 \text{ ft}$, $h = 3 \text{ units} = 15 \text{ ft}$)

$$= 40 \text{ ft} \times 15 \text{ ft} = 600 \text{ ft}^2$$

$$\text{Total area} = 157.08 + 450 + 600 = 1207.08 \text{ ft}^2$$



If students are unsure how to begin, prompt them to use different colors to decompose or “break up” the composite figure into figures that they know.



How do you find the area of a semicircle? (Divide the area of the circle by 2.)



Once students have decomposed the figure into figures they know, prompt students to make connection to what they know about their area.

- What do you know about the area of a circle?
- How is a semicircle related to a circle?
- So, what does that make you think about finding the area of a semicircle?

9.1.5 Wrap-Up: Reflection

Component Overview: Students complete a self-reflection on their level of understanding of the content of this lesson.



9.1.5 Wrap-Up: Reflection

1. Circle the icon that best describes your understanding of each.



I need help.



I got it.



I could help someone.

Answers will vary.

Finding the area of a triangle			
Finding the area of a rectangle			
Finding the area of a trapezoid			
Finding the area of a square			
Finding the area of a rhombus			
Finding the area of a parallelogram			
Finding the area of a circle			
Finding the area of a composite figure			



The information provided from the students here, coupled with what was observed during instruction, can provide information for differentiating instruction in later lessons.



Mark-Up Text

Consider prompting students to incorporate these symbols throughout this lesson to indicate their level of confidence during the learning process. Students can use one of the three symbols to mark each activity, or you can list each activity on a poster and have students draw their symbol for each activity on the way out of the classroom. Then use these symbols as information to differentiate in Lesson 2.



If time permits, choose one of the strands in the left column and prompt students to find a partner with a different symbol chosen in the right column and collaborate. For example, prompt students who selected “I need help” to find a student who selected “I could help someone” and have them ask that student one question. Students who selected “I got it” can choose to either find someone who needs help or find someone who could help them with a specific question.